Week 15 Notes

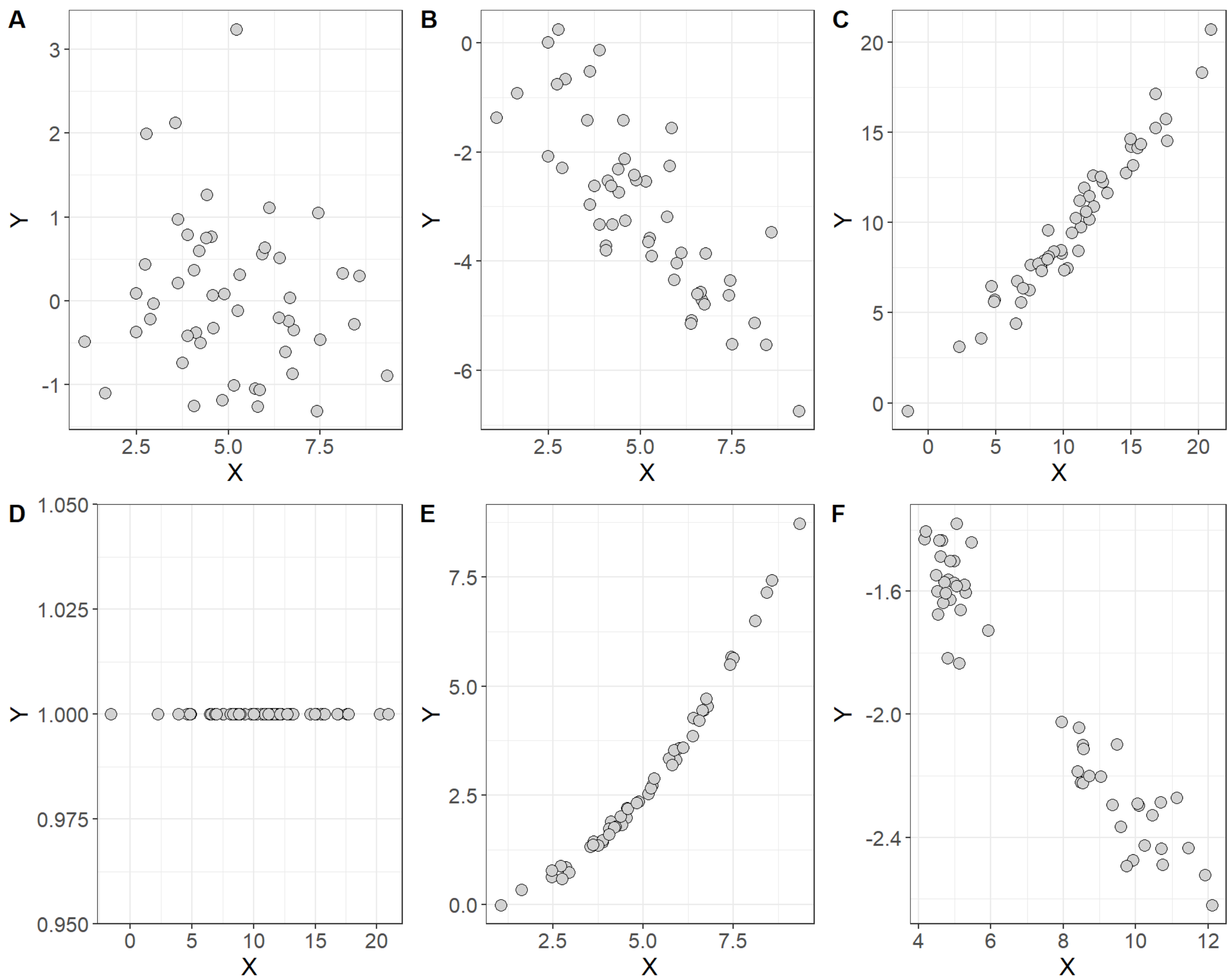
STAT 251 SECTION 01

Lecture 30

In many studies, the goal is to show that changes in one or more explanatory variables actually cause changes in a response variable. The statistical techniques we will describe this week will require us to distinguish between explanatory and response variables. You will often see explanatory variables called the **independent variable** and response variable called the **dependent variable**. The idea behind this language is that the response variable depends on the outcomes of the explanatory variable (if they share a statistical relationship).

Scatterplots and statistical associations

A **scatterplot** is a type of descriptive plot that shows the relationship between two quantitative variables (denoted X and Y) measured on the same individual usually. The horizontal axis is called the x-axis and the vertical axis is called the y-axis. Typically the independent or explanatory variable is represented on the x-axis and the response variable is represented on the y-axis. Consider the four scatterplots below which plot the relationship between the independent variable X and the dependent variable Y:



- Which of these plots indicate a statistical association between the two variables and which do not?

How do you interpret a scatterplot?

- look for an overall patterns of the points in each plot
- You can describe the overall pattern in three terms: **form**, **direction** and **strength**

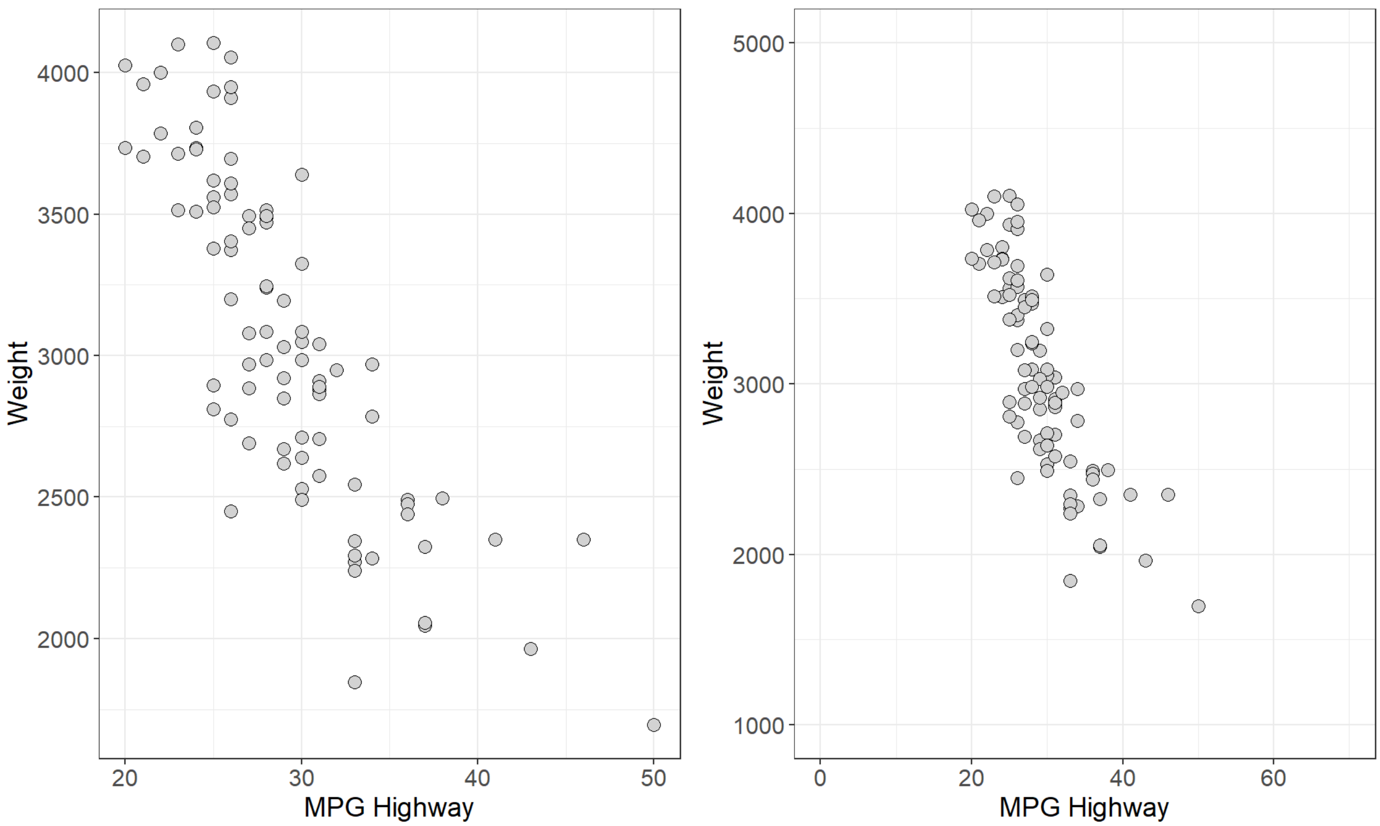
- **form**: do the points form a line, a curve, or a cloud? if the points follow roughly a straight line we say that they are **linear** in form, and if the points follow a curve we would say that they are **non-linear** in form.
- **direction**: which direction do the points trend as you move left to right across the values of the explanatory variable
- **strength**: if there is a pattern, are the points loose and spread out? or are they tightly packed together?

In general, we say that two variables are **positively associated** when larger values of the explanatory variable are paired with larger values of the response variable. Conversely, we say that two variables are **negatively associated** if larger values of the explanatory variable are paired with smaller values of the response.

It may also be important to note if the points form specific clusters as this may indicate a hidden grouping in the data that is not accounted for by the independent variable. When a scatterplot shows distinct clusters it is often most useful to describe the pattern in each cluster.

Measures of association

As we just saw, scatterplots display the form, direction, and strength of the relationship between two quantitative variables. Linear (straight-line) relationships are often of interest because they occur most often. If the points in a scatterplot lie close to a straight line we say that they have a strong linear relationship. However, scatterplots can be misleading when trying to judge the strength of a linear relationship. Consider the two plots below which show the relationship between highway fuel efficiency (in miles per gallon) and weight of cars from the classic 1993 New Car dataset (Lock 1993; Venables and Ripley 2002).



- Which plot shows a stronger linear relationship?

This was a rhetorical question since both plots depict the same data. However, the plot on the right spans a larger range on both the x and y axes. This example demonstrates that the way data is plotted can mislead our perception of the strength of a linear relationship, showing that humans are not always reliable judges in this regard. Therefore, we need to follow our general strategy of following up any graphical display with a numerical measure to help us interpret what we see.

In general when employing numerical measures of association our data generally constitute measurements on two quantitative variables X and Y measured on n individuals

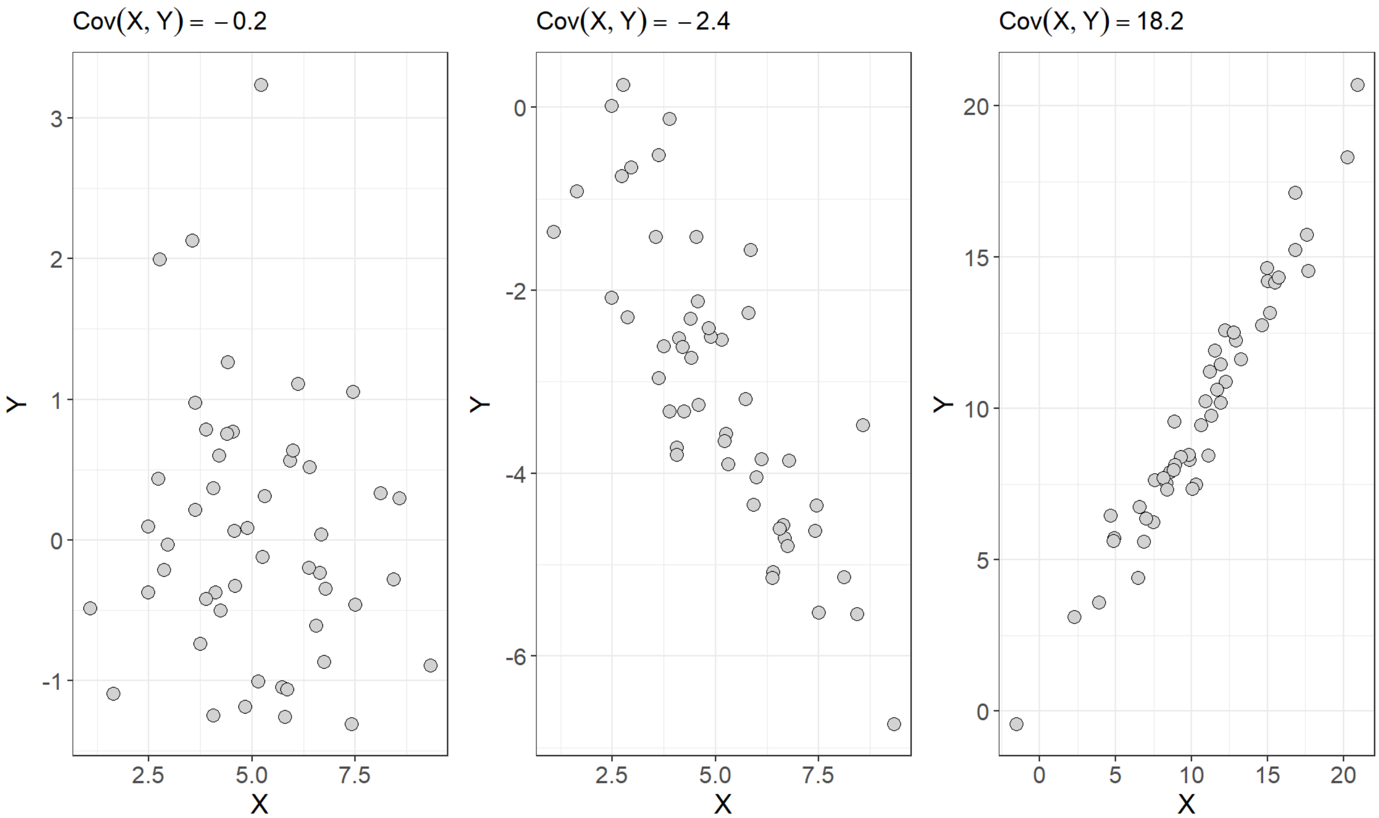
$$X = \{x_1, x_2, x_3, \cdots x_n\}$$

$$Y = \{y_1, y_2, y_3, \cdots y_n\}$$

Each pair of x and y values (x_i, y_i) constitutes a point in a scatterplot of these two variables so that there are n total points in the plot. One measure of association between two variables is called **covariance**. Recall that the sample variance is a numerical measure of the spread of a variable and gives the average size of the squared deviation from the mean. Covariance is the joint variability between two variables X and Y defined as

$$Cov(X, Y) = \sum_i \frac{(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

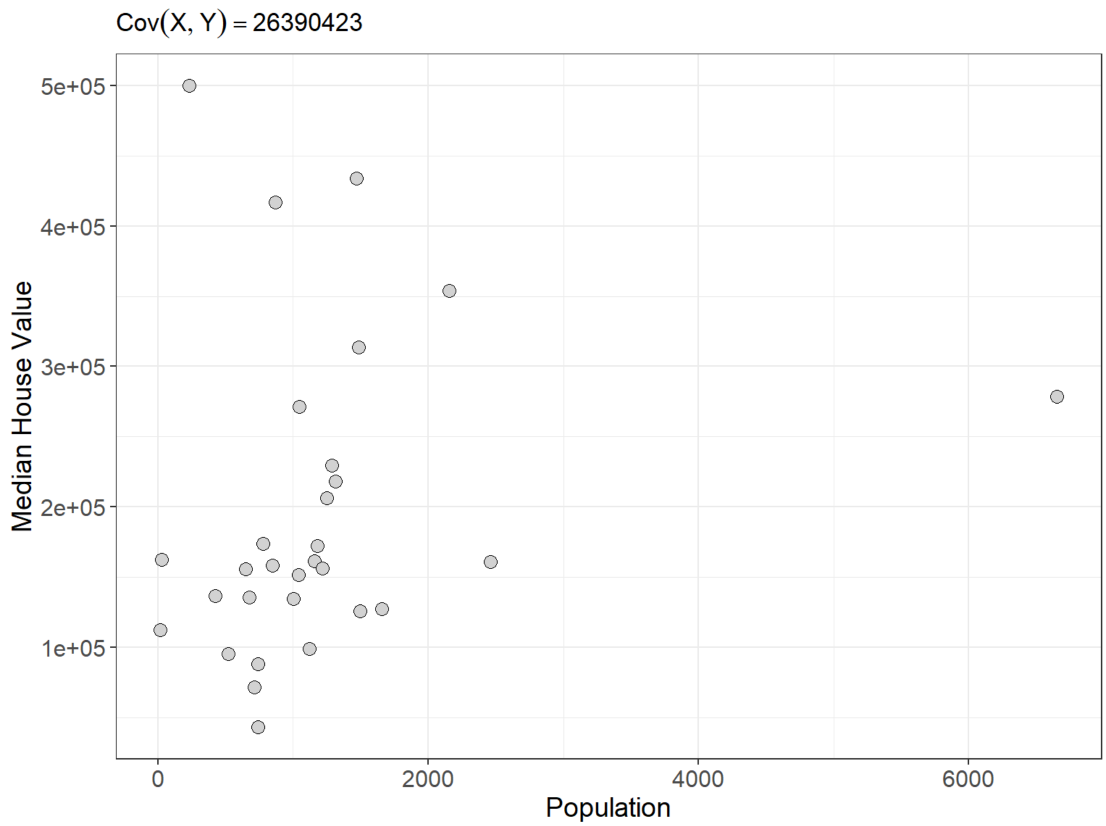
In general, if two variables share a strong positive or strong negative relationship, they will have a large positive or large negative covariance, and if two variables are independent their covariance should be close to zero. Consider the first three scatterplots and the covariances between X and Y from *figure 1* seen earlier



As we can see from above the plots that show a stronger linear relationship are paired with larger covariances for X and Y. However, you may have noticed that covariance depends on the units of X and Y. This makes covariance hard to interpret because one or both of the variables exists on a large scale the covariance will naturally take on large values. Consider the scatterplot below which shows the relationship and covariance between median house value and population for 30 districts in California from 1990 (Géron 2022)

Population of District	Median Home Value
870	416700
648	155500
1660	127100
715	71500
1161	161500
2461	160800
1124	98600

Population of District	Median Home Value
1182	172300
1289	229500
20	112500
742	88200
1498	125700
2156	353800
6652	278300
848	158200
1488	313600
423	136300
1047	271300
1312	218000
1250	206200
28	162500
520	95000
676	135500
1217	156300
1039	151600
1001	134400
232	500001
776	173500
1468	434000
742	43000



Even though the plot doesn't really show a strong relationship between these two variables the covariance we compute is a whopping 26,390,423. Moreover, because our two variables have different units, the units of the covariance we computed are nonsensical. Since covariance is not straight-forward to interpret, it is not typically used to measure the strength of a linear relationship. Instead, a similar measure called **correlation** is the statistical measure that is most often used to determine the strength of the linear relationship between two quantitative variables. We can think of correlation as a type of standardized, unitless covariance. Essentially, correlation is a scaled form of covariance that allows for clearer interpretation of the relationship between two quantitative variables.

For each individual there is a height x_i and a weight y_i. Correlation is defined as

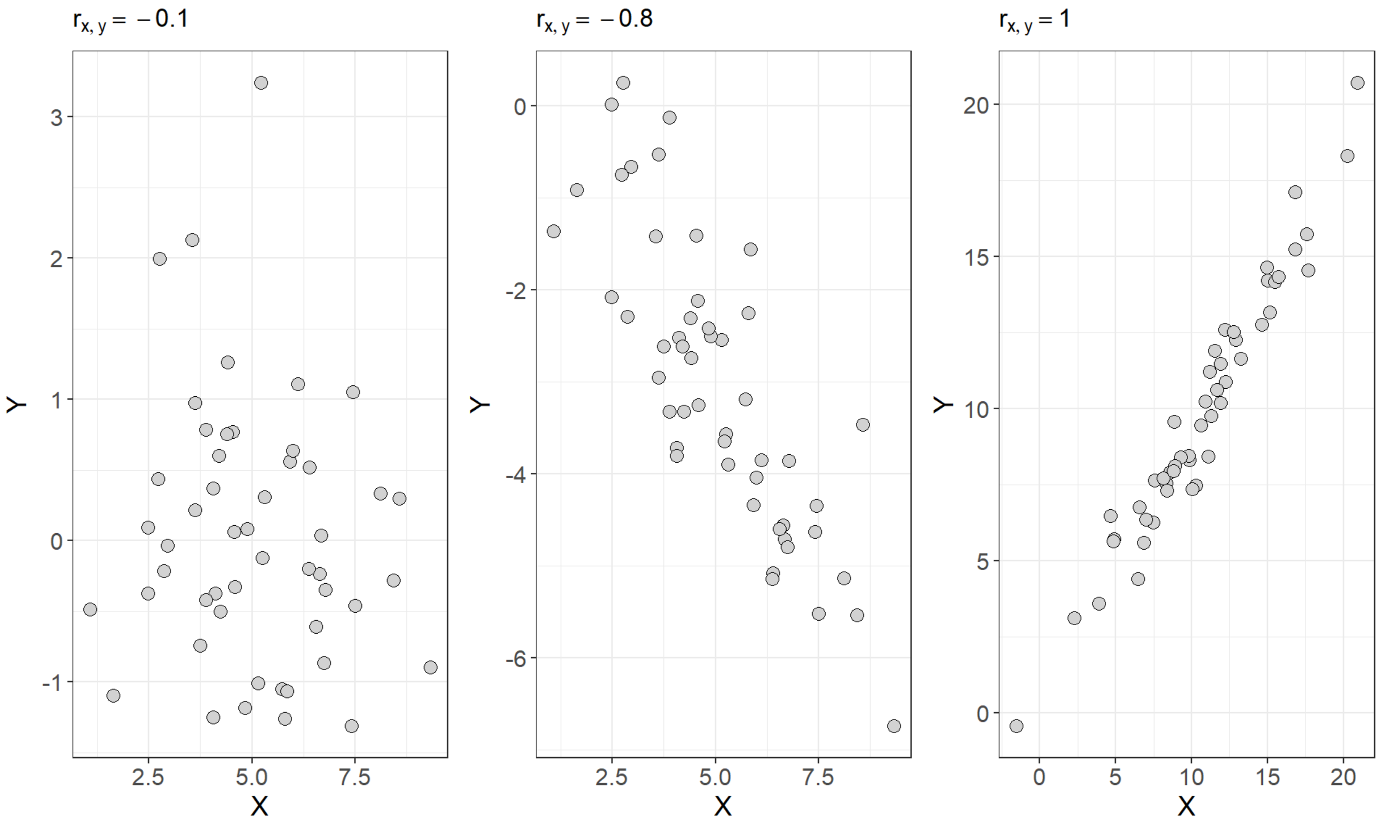
$$r_{x,y} = \frac{1}{n-1}\sum_i \left(\frac{x_i - \bar{x}}{s_x}\right)\left(\frac{y_i - \bar{y}}{s_y}\right) = \frac{Cov(X,Y)}{\sqrt{s_x^2 s_y^2}}$$

The symbol $r_{x,y}$ is used to denote the sample correlation between variables X and Y. From the formula above, we can see that the correlation coefficient $r_{x,y}$ is the covariance of X and Y standardized by the individual standard deviations for each variable. The result is that the r has no units and represents the average of the products of the standardized X and Y values across all n observations. This gives the correlation coefficient several nice properties:

- it ranges in value from -1 to $+1$. Values close to -1 indicate strong negative associations and values close to $+1$ indicate strong positive associations.
- A correlation coefficient of exactly positive or negative one indicates a perfect linear relationship
- values at or close to zero indicate little or no association between the variables.
- Unlike covariance, the value of the sample correlation will NOT change when the units of one or more of the variables are changed. This is because we standardize the deviations to have no units.

It's important to note that correlation can only be used to measure the strength of **linear** relationships between two variables. It cannot accurately describe **non-linear** (curved) relationships. It is also not a resistant measure. Outliers will have a significant effect on the value of the correlation coefficient

In the earlier example comparing population to median home value, the correlation coefficient between these two variables is $r \approx 0.2$. Now consider the first three plots from *figure 1* but this time with correlations plotted for each scatterplot



Computing the covariance or correlation is not an easy task. The formulas are quite involved and infeasible for any large number of observations. Thus it is best to rely on software to easily compute correlations. However, we can get a feel of the mechanics of correlation by computing it for a small number of observations. Try the example below

Try it out: Consider the following dataset which is the height and weight for three adults

Index	Height (inches)	Weight (pounds)
individual 1	69.8	140.9
individual 2	69.8	129
individual 3	65.4	123.2
Sample Mean	68.3	131
Sample SD	2.54	9.02

Let us start by computing the sample covariance for height and weight

$$Cov(\text{Height}, \text{Weight}) = \frac{(69.8-68.3)(140.9-131)}{3-1} + \frac{(69.8-68.3)(129-131)}{3-1} + \frac{(65.4-68.3)(123.2-131)}{3-1}$$
$$\approx 17.24$$

The correlation is given by

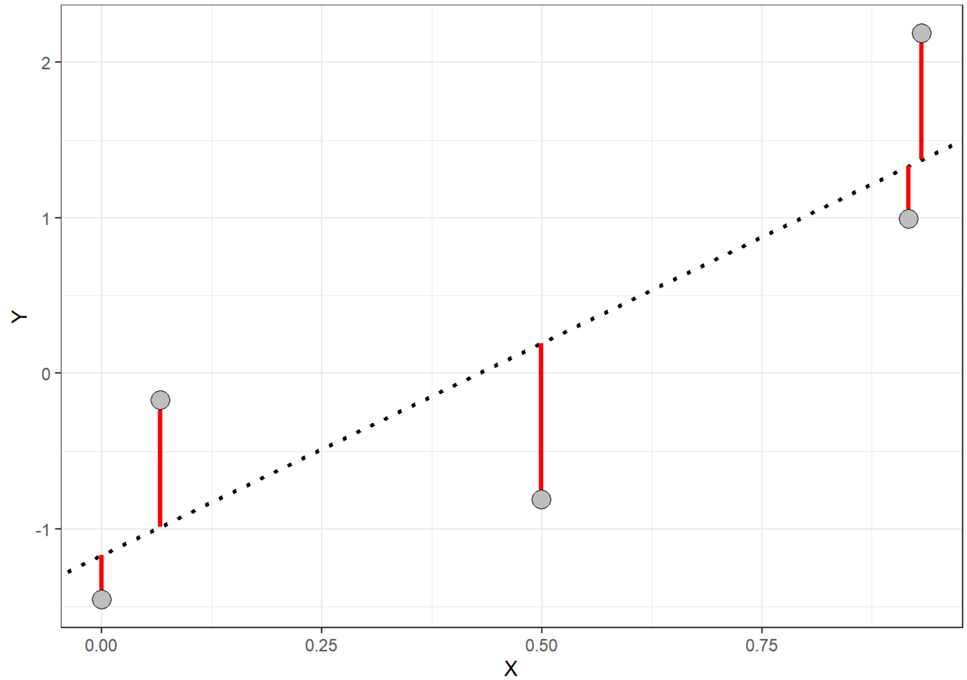
$$r = \frac{17.24}{\sqrt{2.54^2 \cdot 9.02^2}} \approx 0.75$$

Introduction to simple linear regression

Correlation measures the strength of a linear relationship between two variables, but what if you want to go a step further and find the best-fit line that connects those dots? That's where linear regression comes in. It's a statistical technique that finds the straight line that best describes the relationship between two quantitative variables, often visualized on a scatterplot. The most basic version of this technique, where you fit a line to two quantitative variables, is called *simple linear regression*.

When a scatterplot shows a linear pattern we can describe the overall linear relationship by drawing a line through the points. Of course any line that we draw will not touch all of the points. But we would like any line that we draw to come as close to all of the points as possible. A regression line of this sort gives a compact description of how a response variable y changes as an explanatory variable x changes.

It turns out that the line that comes as close as possible to all of the points is the line that minimizes the squared deviations of the **residuals**.



Residuals are the vertical distances (red lines in the plot above) between each point and the regression line. If you think of the regression line as our best guess at the relationship between X and Y, then the residuals represent the error in that guess for each point. The line that "best" fits the data is the one with the smallest average residual—essentially, the one with the least error across all the points in the plot. This line gives us the best model to describe the relationship between the X and Y variables. You can take a look at https://www.geogebra.org/m/xC6zq7Zv (https://www.geogebra.org/m/xC6zq7Zv) to practice fitting the "best" line.

Lecture 31

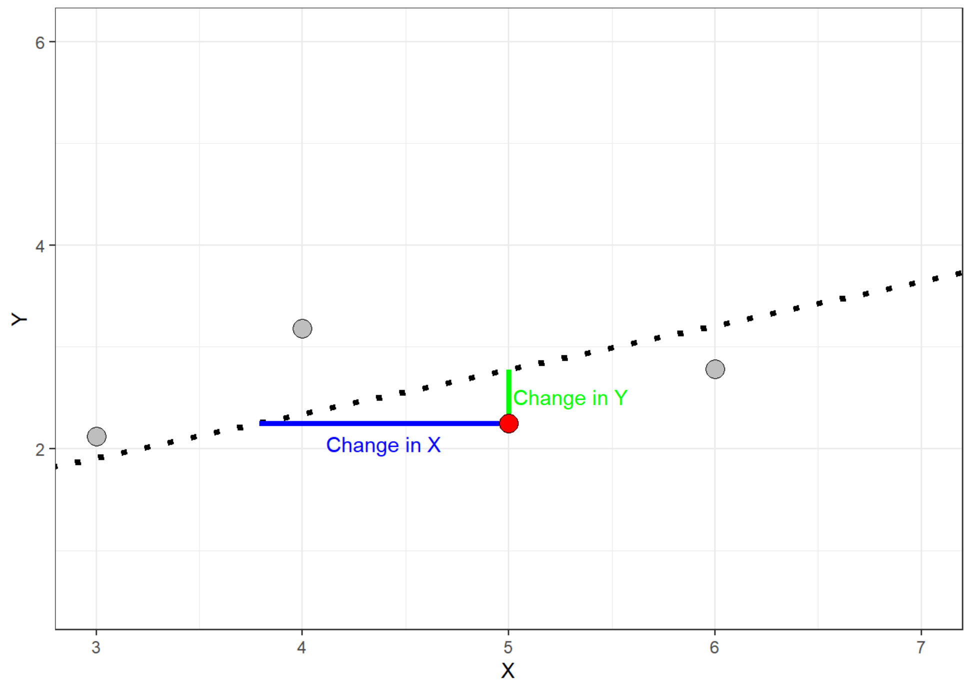
Monday was about *describing* the association between two quantitative variables: a scatterplot to see it, and the correlation coefficient to measure its strength and direction. Today we move from describing that association to *modelling* it. We will write down a straight-line model for the relationship, find the line that fits the data best, use it to predict values of the response, and then measure how much of the variation in the response that line actually explains.

The simple linear model

We learned on Monday that a scatterplot can be used to visualize and check for a **linear** relationship between two quantitative variables. We also learned that straight line fitted through the data points can be used to describe that trend. We will use the notation

$$\hat{y}_i = \alpha + \beta x_i$$

to display this line mathematically, called the **regression line**. The equation above is often referred to as the *prediction equation*. The symbol $\hat{y}_i$ represents the *predicted value* for the i^{th} observation of the dependent variable Y given the i^{th} value of the independent variable x_i. The symbol α is called the y-axis intercept and β is called the slope. The **slope** of a regression line relating X and Y describes how much $\hat{y}$ changes per unit change in X (sometimes called *rise over run*). This is visualized in the plot below.



- When the slope is negative the predicted values decrease as independent variable X increases
- When the slope is positive the predicted values increase as the independent variable X increases
- When the slope is zero (or near zero) the regression line is approximately horizontal.
- The absolute value of the slope gives the magnitude change in Y per unit change of X.
- The slope is dependent on the units of X. A 1-unit increase in X could be a trivial amount or it could be huge.

Finding a regression line and the least-squares estimator

On Monday, we also briefly introduced the idea of a **residual** - the disparity (error) between the values predicted by the regression line $\hat{y}_i$ and the actual values of the response variable y_i. We will use the symbol ϵ_i (called epsilon) to represent the residual for the i^{th} observation of the response variable. We can write the observations of the response variable in terms of the predicted values and the residuals:

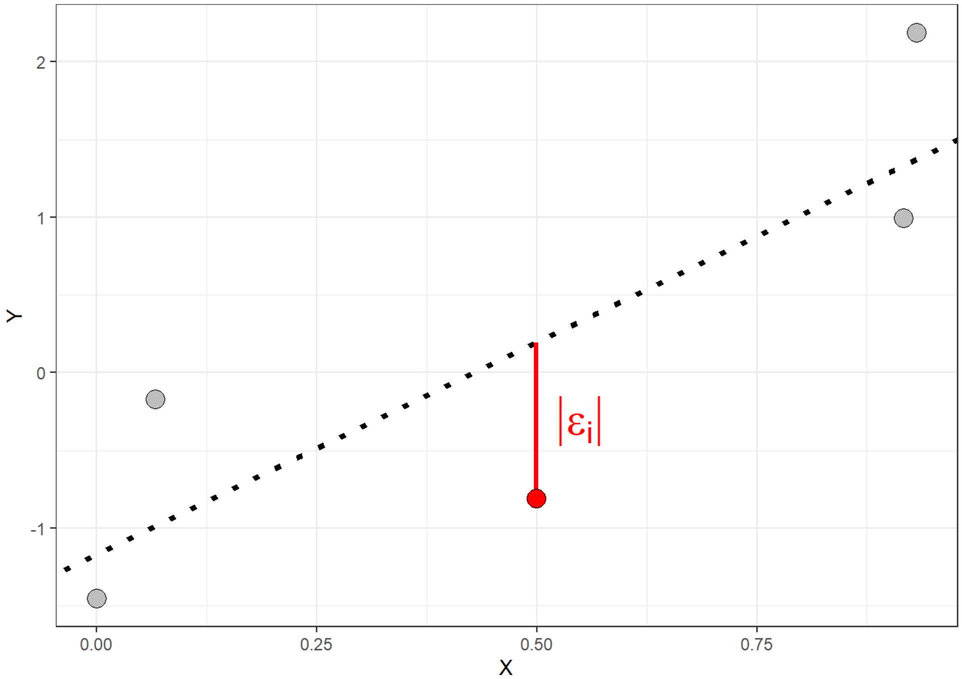
$$y_i = \alpha + \beta x_i + \epsilon_i$$

where

$$\epsilon_i = y_i - \hat{y}_i$$

Thus each observation of the response variable has a residual. Some of the residuals will be positive and some will be negative.

- A positive residual indicates that the actual y_i is greater than predicted value $\hat{y}_i$
- A negative residual indicates that the actual y_i is less than predicted value $\hat{y}_i$
- The smaller the absolute value of the residual the closer the predicted value is to the actual value of the response variable



The length of the red line in the plot above represents the absolute value of the residual for observation y_i. The residuals are all vertical distances because the regression equation predicts values of the response variable.

Recall that the line of "best" fit is the one that is simultaneously as close as possible to all data points. Such a line will have the smallest possible residuals. This process involves compromise as any line we draw can perfectly predict any one point but may gravely over/under predict other points resulting in larger residuals. The actual summary measure used to evaluate and fit a regression line is called the residuals sum of squares (denoted SS_E)

$$SS_E = \sum_i (\hat{y}_i - y_i)^2$$

The measure above describes the variation in the residuals. The line which minimizes the residual sum of squares will have the smallest average residuals. Thus the better the line fits the data, the smaller the residuals will be and the the smaller the SS_E will be.

- The regression line which minimizes the SS_E will have some positive and some negative residuals, but the mean of the residuals will be 0.
- It will pass through the point $(\bar{x}, \bar{y})$. This means the line passes through the center of the data

Even though we will typically rely on software to compute a regression line, the least squares line gives explict formulas for the slope and intercept:

$$\hat{\beta} = r_{x,y}\left(\frac{s_y}{s_x}\right)$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Notice that the slope $\hat{\beta}$ is directly related to the correlation coefficient r

Predicting values of the dependent variable: Batting averages

A regression line allows us to further describe the relationship between two variables that goes beyond just stating the strength of the relationship using the correlation coefficient $r_{X,Y}$. Consider the following data which gives observations of team batting average and average score for teams in the American League of baseball.

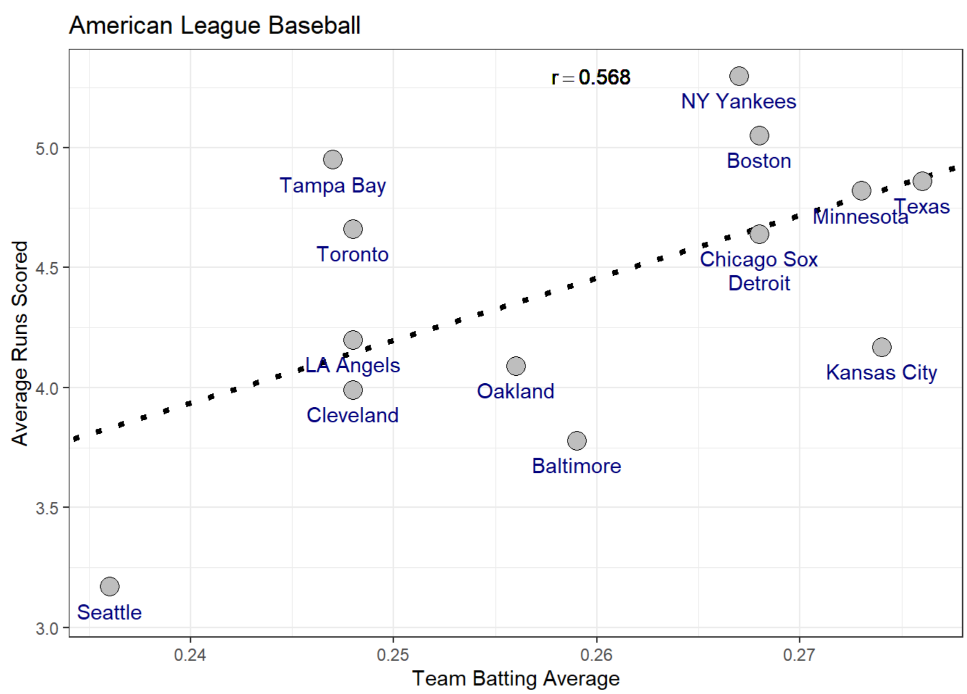
Team	Team Batting Average	Average Runs Scored
NY Yankees	0.27	5.30

Team	Team Batting Average	Average Runs Scored
Boston	0.27	5.05
Tampa Bay	0.25	4.95
Texas	0.28	4.86
Minnesota	0.27	4.82
Toronto	0.25	4.66
Chicago Sox	0.27	4.64
Detroit	0.27	4.64
LA Angels	0.25	4.20
Kansas City	0.27	4.17
Oakland	0.26	4.09
Cleveland	0.25	3.99
Baltimore	0.26	3.78
Seattle	0.24	3.17

The table below summarizes the data.

	Team Batting Average	Average Runs Scored
Sample Standard Deviation	0.013	0.577
Sample Mean	0.260	4.451

Now consider fitting a line to estimate the relationship between team batting average (independent variable) and average runs scored per game (dependent variable). The plot below gives the least-squares regression line relating these two variables. Based on the plot, is team batting average a good predictor of the average number of runs a team will score ?



- Use the plot and the summary information to compute the estimates for the slope and intercept

Using the summary table ($\bar{x} = 0.260$, $s_x = 0.013$, $\bar{y} = 4.451$, $s_y = 0.577$) and the correlation $r = 0.568$ from the plot:

$$\hat{\beta} = r\left(\frac{s_y}{s_x}\right) = 0.568\left(\frac{0.577}{0.013}\right) \approx 25.2$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 4.451 - 25.2(0.260) \approx -2.10$$

- Using the prediction equation, how many runs would we expect a team with a batting average of 0.25 to score?

$$\hat{y} = -2.10 + 25.2(0.25) \approx 4.20 \text{ runs}$$

- Is there a significant association between batting average and runs scored? how can you tell?

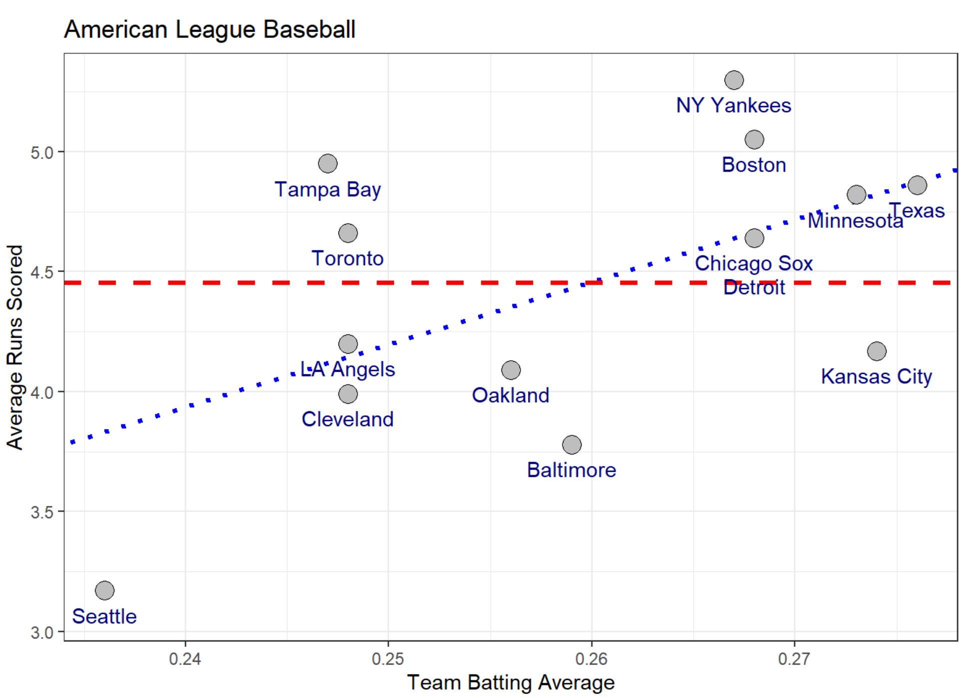
The scatterplot shows a moderate positive linear trend with $r \approx 0.568$, so $r^2 \approx 0.32$ - about 32% of the variation in runs scored is explained by batting average. To confirm the association is statistically significant we would test $H_0 : \beta = 0$ against $H_A : \beta \neq 0$; the clear upward trend and moderate correlation indicate the slope is significantly greater than zero.

Important Note on Regression Predictions: When applying a regression line to make predictions, it is crucial to avoid extrapolating beyond the range of values in the explanatory variable used to create the model. Extrapolation involves estimating outcomes for values of the predictor variable (X) that fall outside the range observed in the data. This practice is discouraged because the linear relationship between X and Y established by the model might not hold true beyond the observed data.

In other words, regression models are built to represent trends within the scope of the available data. Once you go beyond the minimum or maximum observed values of the explanatory variable, you enter a zone where the relationship between X and Y could be different. This uncertainty can lead to inaccurate or misleading predictions. Therefore, always ensure that predictions are made within the range of data used to fit the model to maintain reliability and accuracy.

Coefficient of determination r^2

When we predict a value of Y, why should we use the regression line? We could instead use the center of Y's distribution as our prediction for Y such as the sample mean $\bar{y}$. One reason is that if the explanatory variable X and response Y have an approximate linear relationship then the prediction equation $\hat{y}_i = \alpha + \beta x_i$ will give more accurate predictions than simply using the sample mean $\bar{y}$. Consider again the example of teams in the American League for Major League Baseball



The red line gives the predictions using the sample mean and the blue line is the using the least squares regression line. In general, the stronger the linear relationship between the independent and dependent variables, the better the predictions with the regression line will become.

To judge how much better the predictions are for the regression model we can compare the residuals of the model fitting the regression line (blue line above) to the model fitting the sample mean (red line above).

The residuals for the model using the sample mean is given by

$$SS_T = \sum_i (y_i - \bar{y})^2$$

called the *total sum of squares* or SS_T because it represents the total variation in the response variable. The residuals for the regression model are given by

$$SS_E = \sum_i (\hat{y}_i - y_i)^2$$

The difference between the two is called the *regression sum of squares* or SS_R

$$SS_R = SS_T - SS_E = \sum_i (\hat{y} - \bar{y})^2$$

Notice that by rearranging the equation above, the total variation in the response can be partitioned in to two parts: the variation explained by the predictor (i.e SS_R) and the variation not explained by the predictor (i.e SS_E).

$$SS_T = SS_E + SS_R$$

By comparing the SS_T and SS_R, we can determine how much the regression model improves our predictions over the model fitting the sample mean. This measure is called the **coefficient of determination** denoted r^2 because it is the square of the correlation coefficient r.

$$r^2 = \frac{SS_R}{SS_T} = \frac{\sum_i (\hat{y} - \bar{y})^2}{\sum_i (y_i - \bar{y})^2}$$

The r-squared value is expressed as a percentage and tells us how much the overall variation in the response can be explained by the linear regression model. For the American League example we have a total sum of squares of

$$SS_T = 4.323$$

and a regression sum of squares of

$$SS_R = 1.396$$

which gives a coefficient of determination of

$$r^2 = \frac{1.396}{4.323} = 0.323 \ \text{ or } \ 32.3\%$$

Thus team batting average explains roughly 32% of the variation in the average number of runs scored by teams in the American League of Major League Baseball.

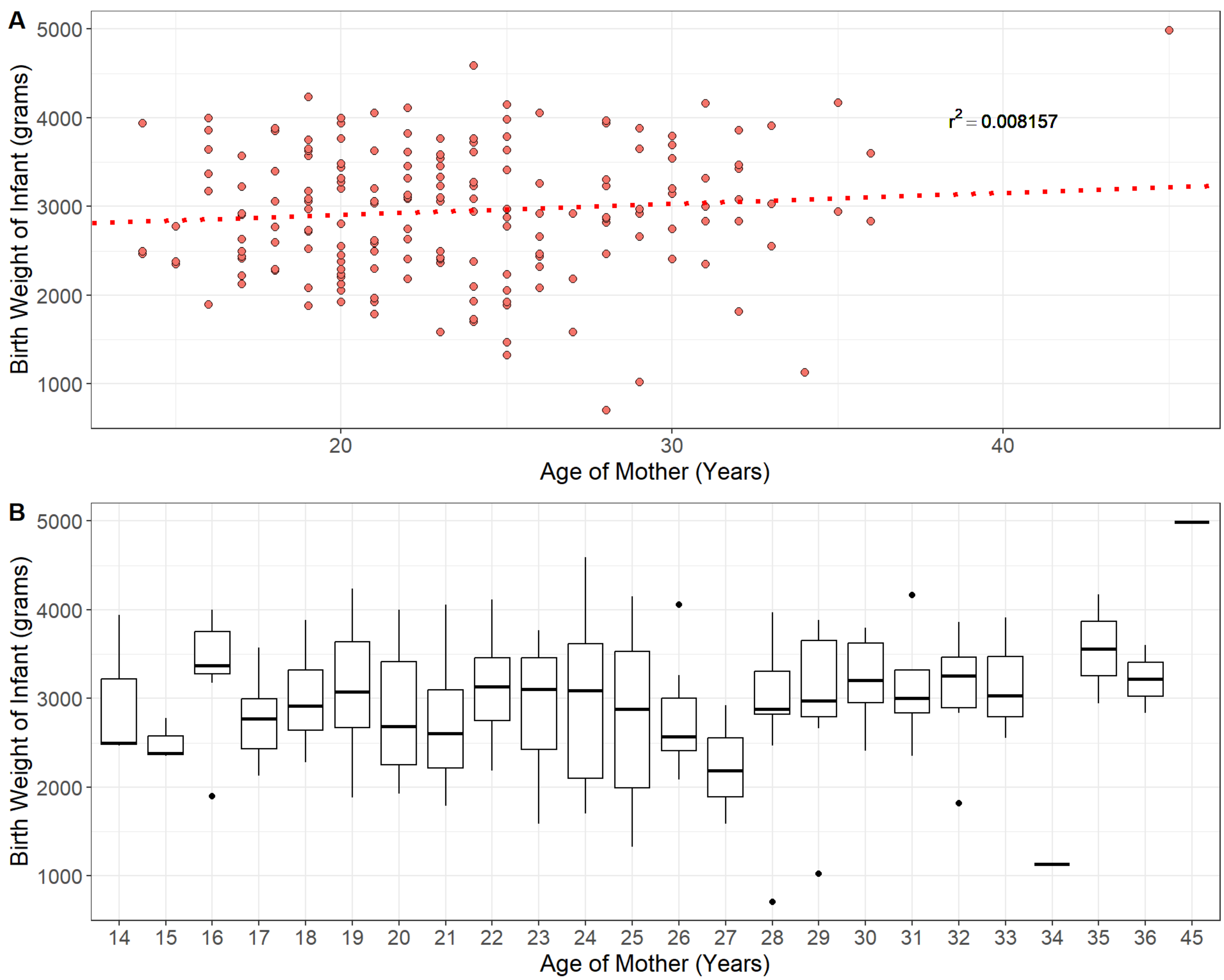
Lecture 32

So far the regression line has been a *description* of the sample: we found the least-squares line, used it to predict, and measured its fit with r^2. But the slope we computed is a statistic, and like any statistic it would change if we collected a different sample. Today we treat regression as *inference*. We ask whether the association we see is strong enough to conclude that a relationship exists in the population, we test the slope, we look at a second test of the same question based on variances, and we finish by checking the assumptions that make any of those conclusions trustworthy.

Regression relates the mean of Y to X

The prediction equation $\hat{y}_i = \alpha + \beta x_i$ yields the predicted value the response Y for a given value of the independent variable X. However, we don't expect all observations of Y that depend on a given value of X to have the same value because naturally there is variation among individuals in the population. For example, consider the linear regression below which relates the age of 189 mothers in the U.S to the birth weight of their infants (Venables and Ripley 2002).

Individual	Age of Mother (Years)	Infant Birth Weight (grams)
individual 1	19	2523
individual 2	33	2551
individual 3	20	2557
individual 4	21	2594
individual 5	18	2600
⋮	⋮	⋮
individual 189	21	2495



The prediction equation for the regression model fit to the birthweight data is given by

$$\hat{y} = 2655.7 + 12.43x_i$$

Now consider the predicted birth weight of an infant born to a mother of 21 years of age: $\hat{y} = 2916.7$ grams. We do not expect all infants born to mothers of 21 years of age have the same weight of 2916.7 grams. One mother may have an infant born at 3200 grams and another at 2500 grams and so on. As we can see from panel $\mathbf{B}$ in the plot above, for each possible age of the mother, there is a distribution of infant birth weights corresponding to all mothers of that age. Thus we can think of the predicted value $\hat{y} = 2916.7$ as estimating the mean birth weight of all mothers aged $x = 21$ years.

The **population regression equation** relates the predictor x to the means of Y in the population.

$$\mu_y = \alpha + \beta x$$

where α, and β are population parameters that are unknown. The quantity μ_y is the population mean of y for a given value of x.

Statistical inference for linear regression

A linear regression model:

- specifies the linear relationship between two quantitative variables
- uses a straight line to relate the means of Y across specific values of the predictor variable x

The **population regression equation** contains two population parameters α the intercept and β the slope which are important for understanding the linear relationship between the predictor and outcome variables. We can conduct statistic inference on these two population parameters to place bounds on their values using confidence intervals or conduct hypothesis tests about specific values. However, as always, statistical inference comes with certain assumptions about the model which we will list below:

Statistical inference for the least-squares simple linear regression is based on the model

$$y_i = \alpha + \beta x_i + \epsilon_i$$

the key assumptions of this model mostly concern the behavior of the residuals ϵ_i:

- **Linearity** - the expected value (i.e average value) of the residuals is 0 for all values of x: $E[\epsilon_i|x] = 0$
- **Constant Variance** - the variance of the residuals is the same regardless of the value of the predictor x: $V[\epsilon_i|x_i] = \sigma_\epsilon^2$
- **Normality** - residuals are normally distributed: $\epsilon_i \sim N(0, \sigma_\epsilon^2)$
- **Independence** - the observations are sampled independently: any pair of errors (ϵ_i, ϵ_j) are independent
- **Fixed** X - the values of the predictor X are fixed values or are measured without error and are independent of the residuals
- X **is not invariant** - If the explanatory variable is fixed, then its values cannot all be the same. (i.e x cannot be constant)

Under these assumptions the population parameters α and β are normally distributed. We are generally more concerned with the slope than the intercept. Therefore, the $100(1-a)\%$ confidence interval (a represents the significance level) for the slope is given by

$$\beta = \hat{\beta} \pm t_{1-a/2} SE(\hat{\beta})$$

In the equation above the critical value $t_{1-a/2}$ follows a t-distribution with $n-2$ degrees of freedom.

the standard error of the sample slope is given by

$$SE(\hat{\beta}) = \sqrt{\frac{s_\epsilon^2}{(n-1)s_x^2}}$$

$$= \sqrt{\frac{\frac{1}{n-2}\sum_i \hat{\epsilon}_i^{\,2}}{\sum_i (x_i - \bar{x})^2}}$$

A hypothesis test on the population slope β is usually conducted under the null hypothesis $H_0 : \beta = 0$ or "no linear relationship between X and Y". The test is generally two-sided with alternative hypothesis $H_A : \beta \neq 0$. The test statistic is the standardized estimate (with $\beta_0 = 0$ under the null)

$$t_0 = \frac{\hat{\beta} - \beta_0}{SE(\hat{\beta})} \sim t(n-2)$$

The calculations at this point are quite involved and it is best to rely on software to conduct inference on regression coefficients. Consider, again, the example of a regression model relating the age of mothers to the birth weight of their infants. The programming software R makes fitting linear models quite easy:

```
set.seed(123)

# The data come from the R-package MASS
data = MASS::birthwt

#fit a linear model using the "lm()" function: model = lm(y ~ x, data = dataframe)
model = lm(bwt ~ age, data = data)
print(model)
```

```
##
## Call:
## lm(formula = bwt ~ age, data = data)
##
## Coefficients:
## (Intercept)          age
##     2655.74        12.43
```

```
#R has a function called summary which nicely displays all of the information:
summary(model)
```

```
##
## Call:
## lm(formula = bwt ~ age, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2294.78  -517.63    10.51   530.80  1774.92
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2655.74     238.86   11.12   <2e-16 ***
## age            12.43      10.02    1.24    0.216
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 728.2 on 187 degrees of freedom
## Multiple R-squared:  0.008157,	Adjusted R-squared:  0.002853
## F-statistic: 1.538 on 1 and 187 DF,  p-value: 0.2165
```

In the output from R above, we see that the estimate of the slope is $\hat{\beta} = 12.43$ with a standard error of $SE(\hat{\beta}) = 10.02$. Recall from earlier that the data consist of 189 mothers and their infant birth weights. Using the estimate and standard error, we can easily construct a 95% confidence interval for the slope

$$12.43 \pm 1.97(10.02) \approx \big[-7.31, 32.17\big]$$

Our confidence interval for the slope overlaps zero which indicates that there is not a strong linear relationship between age of the mother and infant birth weight. The output from above also gives the test statistic and p-value for the hypothesis tests on each coefficient under the null that their values are zero i.e $H_0 : \beta = 0$. The test statistic for the slope in our example is given by

$$t_0 = \frac{12.43}{10.02} = 1.24$$

You can confirm the p-value is ≈ 0.22 using the t-table on the resources tab of the course website. As we can see from the R output, the hypothesis test also confirms that age of mother and infant birth weight are not significantly associated.

An alternative way to test association

Consider the following dataset containing information about 1,143 social media ads from an anonymous company, detailing the number of times each ad was viewed and the total number of customers who made a purchase after viewing each ad (Mohammadi and Burke 2023).

Index	Total Customers Purchasing Product	Number of Times Ad Shown
1	1	7350
2	0	17861
3	0	693
4	0	4259
5	1	4133
6	1	1915
7	0	15615
8	1	10951
9	0	2355
10	0	9502
$\vdots$	$\vdots$	$\vdots$
1142	2	790253
1143	2	513161

Let's consider a linear regression model where the response variable is the number of customers who purchased the product after viewing an ad, and the predictor variable is the number of times each ad was viewed. The least-squares regression output for this model is presented below.

```r
#fit a linear model using the "lm()" function: model = lm(y ~ x, data = dataframe)
model = lm(`Customers Purchasing Product` ~ `Number of Times Ad Shown`, data = data)

#R has a function called summary which nicely displays all of the information:
summary(model)
```

```
##
## Call:
## lm(formula = `Customers Purchasing Product` ~ `Number of Times Ad Shown`,
##     data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.6645 -0.4463 -0.2425  0.6678 12.8558
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                2.341e-01  4.368e-02    5.36 1.01e-07 ***
## `Number of Times Ad Shown` 3.802e-06  1.199e-07   31.69  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.268 on 1141 degrees of freedom
## Multiple R-squared:  0.4682, Adjusted R-squared:  0.4677
## F-statistic:  1005 on 1 and 1141 DF,  p-value: < 2.2e-16
```

The last line of the output in the example above reports something called an F-statistic. This statistic comes from an alternative test for the association between the response and predictor variable. This alternative hypothesis test is based on the relative variation explained by the predictor, similar to the r^2 coefficient, called an **Analysis of Variance** or **ANOVA**. The F-statistic is a test statistic based on the ratio of the *average squared residual* for the regression to *average squared residual for the error*. Recall the relationship between the total variation in the response, the regression and the residuals

$$SS_T = SS_E + SS_R$$

$$\sum_i (y_i - \bar{y})^2 = \sum_i (y_i - \hat{y}_i)^2 + \sum_i (\hat{y}_i - \bar{y})^2$$

The F-statistic compares the relative size of the *mean square* of the regression

$$MS_R = \frac{SS_R}{k}$$

to the mean square of the residuals

$$MS_E = \frac{SS_E}{n-k-1}$$

where k and $n-k-1$ are the respective degrees of freedom for each estimated sum of squares. In simple linear regression between two *quantitative* variables k is always 1. This gives the F-statistic as

$$\begin{aligned} F_0 &= \frac{\text{Mean square for regression}}{\text{Mean square of error}} \\ &= \frac{SS_R/k}{SS_E/(n-k-1)} \sim F(k, n-k-1) \end{aligned}$$

The above statistic is a ratio of two variances, which follows a special type of distribution called an F-distribution, named by George Snedecor in honor of Sir Ronald Fisher, one of the founders of modern statistics. The F distribution has two degrees of freedom parameters, one for the variance in the numerator and one for the variance in the denominator. Similar to the coefficient of determination, ANOVA evaluates how much of the total variation in the response variable is attributed to the regression model. If there is a linear relationship between the predictor and the response variable, it implies that the predictor accounts for a significant portion of the response's variance, leading to a large F-test statistic.

For simple linear regression between two quantitative variables, the F-test is always

$$= \frac{SS_R}{SS_E/(n-2)} \sim F(1, n-2)$$

which is exactly the squared value of the t statistic from the t-test on the slope. The results of an ANOVA are usually displayed in an ANOVA table:

Term	Sum of Squares	Degrees of Fredom	Mean Square	F-statistic	p-value
Predictor	$SS_R = \sum(\hat{y} - \bar{y})^2$	k	SS_R/k	$F_0 = \frac{(SS_R/k)}{SS_E/(n-k-1)}$	$P(F > F_0 \mid H_0)$
Error	$SS_E = \sum(y - \hat{y})^2$	$n-k-1$	$SS_E/(n-k-1)$		
Total	$SS_T = \sum(y - \bar{y})^2$	$n-1$			

Example: Consider again the advertising data from earlier. Below is the ANOVA table for the regression relating the number of customers who purchases a product and the number of times each ad was shown.

```
set.seed(123)


#fit a linear model using the "lm()" function: model = lm(y ~ x, data = dataframe)
model = lm(`Customers Purchasing Product` ~ `Number of Times Ad Shown`, data = data)

#R has a function called "anova" which nicely displays all of the information:
anova(model)
```

```
## Analysis of Variance Table
##
## Response: Customers Purchasing Product
##                              Df Sum Sq Mean Sq F value    Pr(>F)
## `Number of Times Ad Shown`    1 1614.5 1614.53  1004.5 < 2.2e-16 ***
## Residuals                  1141 1833.9    1.61
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

An Application Example

Example: Consider the following dataset which records the nutrition information for nearly 80 popular brands of cereal (Mohammadi and Burke 2023).

Brand	Manuf.	Type	Calories	Protein (g)	Fat (g)	Sodium (mg)	Fiber (g)	Carbo. (g)	Added Sugars (g)	Potassium (g)	Vitamins	Shelf Number	Cups Per Serv.	Customer Rating
100% Bran	N	cold	70	4	1	130	10.0	5.0	6	280	25	3	0.33	68.40
100% Natural Bran	Q	cold	120	3	5	15	2.0	8.0	8	135	0	3	1.00	33.98
All-Bran	K	cold	70	4	1	260	9.0	7.0	5	320	25	3	0.33	59.43
All-Bran with Extra Fiber	K	cold	50	4	0	140	14.0	8.0	0	330	25	3	0.50	93.70
Almond Delight	R	cold	110	2	2	200	1.0	14.0	8	-1	25	3	0.75	34.38

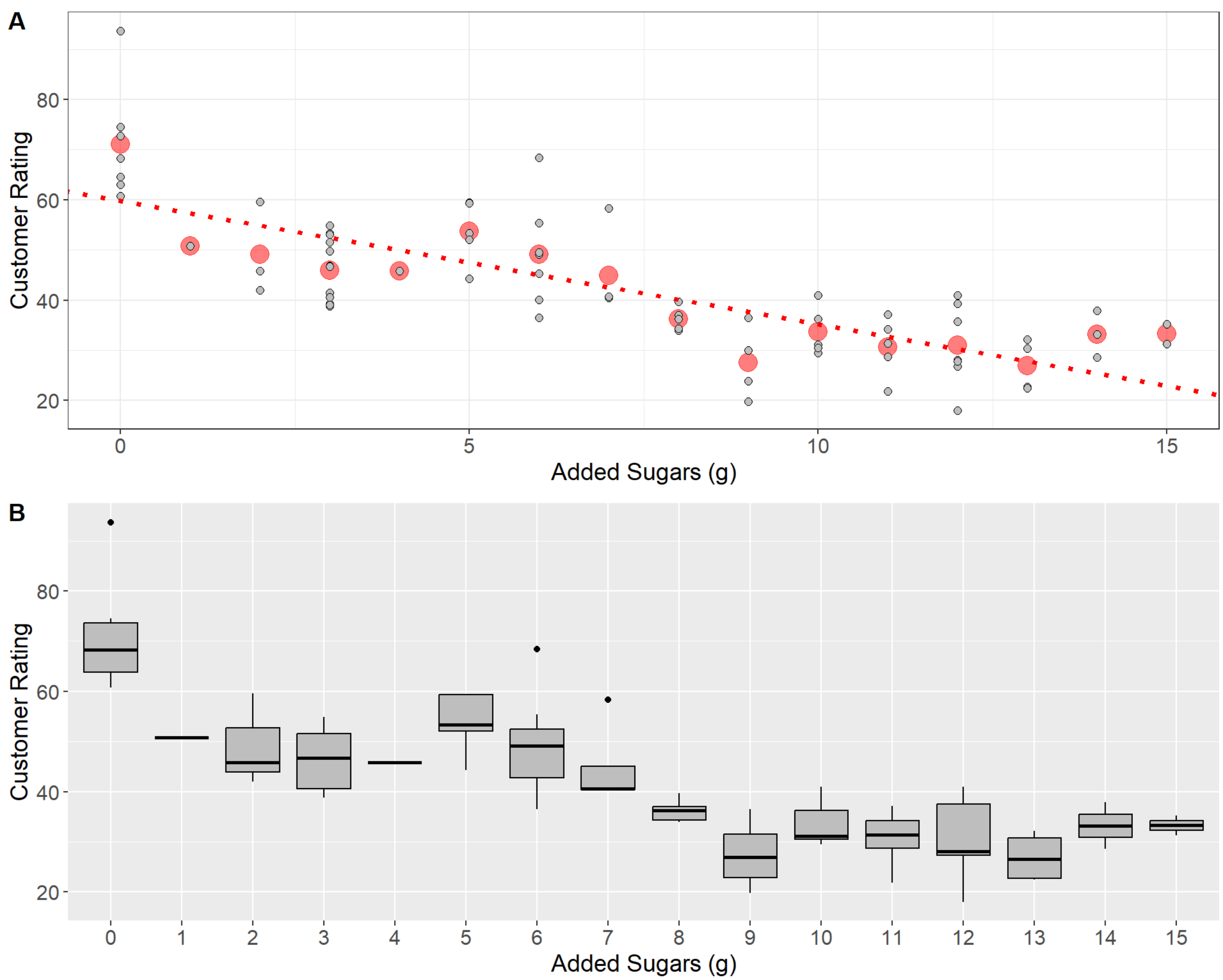
Brand	Manuf.	Type	Calories	Protein (g)	Fat (g)	Sodium (mg)	Fiber (g)	Carbo. (g)	Added Sugars (g)	Potassium (g)	Vitamins	Shelf Number	Cups Per Serv.	Customer Rating
Apple Cinnamon Cheerios	G	cold	110	2	2	180	1.5	10.5	10	70	25	1	0.75	29.51
Apple Jacks	K	cold	110	2	0	125	1.0	11.0	14	30	25	2	1.00	33.17
Basic 4	G	cold	130	3	2	210	2.0	18.0	8	100	25	3	0.75	37.04
Bran Chex	R	cold	90	2	1	200	4.0	15.0	6	125	25	1	0.67	49.12
Bran Flakes	P	cold	90	3	0	210	5.0	13.0	5	190	25	3	0.67	53.31
Cap'n'Crunch	Q	cold	120	1	2	220	0.0	12.0	12	35	25	2	0.75	18.04
Cheerios	G	cold	110	6	2	290	2.0	17.0	1	105	25	1	1.25	50.76
Cinnamon Toast Crunch	G	cold	120	1	3	210	0.0	13.0	9	45	25	2	0.75	19.82
Clusters	G	cold	110	3	2	140	2.0	13.0	7	105	25	3	0.50	40.40
Cocoa Puffs	G	cold	110	1	1	180	0.0	12.0	13	55	25	2	1.00	22.74
Corn Chex	R	cold	110	2	0	280	0.0	22.0	3	25	25	1	1.00	41.45
Corn Flakes	K	cold	100	2	0	290	1.0	21.0	2	35	25	1	1.00	45.86
Corn Pops	K	cold	110	1	0	90	1.0	13.0	12	20	25	2	1.00	35.78
Count Chocula	G	cold	110	1	1	180	0.0	12.0	13	65	25	2	1.00	22.40
Cracklin' Oat Bran	K	cold	110	3	3	140	4.0	10.0	7	160	25	3	0.50	40.45
Cream of Wheat (Quick)	N	hot	100	3	0	80	1.0	21.0	0	-1	0	2	1.00	64.53
Crispix	K	cold	110	2	0	220	1.0	21.0	3	30	25	3	1.00	46.90
Crispy Wheat & Raisins	G	cold	100	2	1	140	2.0	11.0	10	120	25	3	0.75	36.18
Double Chex	R	cold	100	2	0	190	1.0	18.0	5	80	25	3	0.75	44.33
Froot Loops	K	cold	110	2	1	125	1.0	11.0	13	30	25	2	1.00	32.21
Frosted Flakes	K	cold	110	1	0	200	1.0	14.0	11	25	25	1	0.75	31.44
Frosted Mini-Wheats	K	cold	100	3	0	0	3.0	14.0	7	100	25	2	0.80	58.35
Fruit & Fibre Dates; Walnuts; and Oats	P	cold	120	3	2	160	5.0	12.0	10	200	25	3	0.67	40.92

Brand	Manuf.	Type	Calories	Protein (g)	Fat (g)	Sodium (mg)	Fiber (g)	Carbo. (g)	Added Sugars (g)	Potassium (g)	Vitamins	Shelf Number	Cups Per Serv.	Customer Rating
Fruitful Bran	K	cold	120	3	0	240	5.0	14.0	12	190	25	3	0.67	41.02
Fruity Pebbles	P	cold	110	1	1	135	0.0	13.0	12	25	25	2	0.75	28.03
Golden Crisp	P	cold	100	2	0	45	0.0	11.0	15	40	25	1	0.88	35.25
Golden Grahams	G	cold	110	1	1	280	0.0	15.0	9	45	25	2	0.75	23.80
Grape Nuts Flakes	P	cold	100	3	1	140	3.0	15.0	5	85	25	3	0.88	52.08
Grape-Nuts	P	cold	110	3	0	170	3.0	17.0	3	90	25	3	0.25	53.37
Great Grains Pecan	P	cold	120	3	3	75	3.0	13.0	4	100	25	3	0.33	45.81
Honey Graham Ohs	Q	cold	120	1	2	220	1.0	12.0	11	45	25	2	1.00	21.87
Honey Nut Cheerios	G	cold	110	3	1	250	1.5	11.5	10	90	25	1	0.75	31.07
Honey-comb	P	cold	110	1	0	180	0.0	14.0	11	35	25	1	1.33	28.74
Just Right Crunchy Nuggets	K	cold	110	2	1	170	1.0	17.0	6	60	100	3	1.00	36.52
Just Right Fruit & Nut	K	cold	140	3	1	170	2.0	20.0	9	95	100	3	0.75	36.47
Kix	G	cold	110	2	1	260	0.0	21.0	3	40	25	2	1.50	39.24
Life	Q	cold	100	4	2	150	2.0	12.0	6	95	25	2	0.67	45.33
Lucky Charms	G	cold	110	2	1	180	0.0	12.0	12	55	25	2	1.00	26.73
Maypo	A	hot	100	4	1	0	0.0	16.0	3	95	25	2	1.00	54.85
Muesli Raisins; Dates; & Almonds	R	cold	150	4	3	95	3.0	16.0	11	170	25	3	1.00	37.14
Muesli Raisins; Peaches; & Pecans	R	cold	150	4	3	150	3.0	16.0	11	170	25	3	1.00	34.14
Mueslix Crispy Blend	K	cold	160	3	2	150	3.0	17.0	13	160	25	3	0.67	30.31
Multi-Grain Cheerios	G	cold	100	2	1	220	2.0	15.0	6	90	25	1	1.00	40.11

Brand	Manuf.	Type	Calories	Protein (g)	Fat (g)	Sodium (mg)	Fiber (g)	Carbo. (g)	Added Sugars (g)	Potassium (g)	Vitamins	Shelf Number	Cups Per Serv.	Customer Rating
Nut&Honey Crunch	K	cold	120	2	1	190	0.0	15.0	9	40	25	2	0.67	29.92
Nutri-Grain Almond-Raisin	K	cold	140	3	2	220	3.0	21.0	7	130	25	3	0.67	40.69
Nutri-grain Wheat	K	cold	90	3	0	170	3.0	18.0	2	90	25	3	1.00	59.64
Oatmeal Raisin Crisp	G	cold	130	3	2	170	1.5	13.5	10	120	25	3	0.50	30.45
Post Nat. Raisin Bran	P	cold	120	3	1	200	6.0	11.0	14	260	25	3	0.67	37.84
Product 19	K	cold	100	3	0	320	1.0	20.0	3	45	100	3	1.00	41.50
Puffed Rice	Q	cold	50	1	0	0	0.0	13.0	0	15	0	3	1.00	60.76
Puffed Wheat	Q	cold	50	2	0	0	1.0	10.0	0	50	0	3	1.00	63.01
Quaker Oat Squares	Q	cold	100	4	1	135	2.0	14.0	6	110	25	3	0.50	49.51
Raisin Bran	K	cold	120	3	1	210	5.0	14.0	12	240	25	2	0.75	39.26
Raisin Nut Bran	G	cold	100	3	2	140	2.5	10.5	8	140	25	3	0.50	39.70
Raisin Squares	K	cold	90	2	0	0	2.0	15.0	6	110	25	3	0.50	55.33
Rice Chex	R	cold	110	1	0	240	0.0	23.0	2	30	25	1	1.13	42.00
Rice Krispies	K	cold	110	2	0	290	0.0	22.0	3	35	25	1	1.00	40.56
Shredded Wheat	N	cold	80	2	0	0	3.0	16.0	0	95	0	1	1.00	68.24
Shredded Wheat ’n’Bran	N	cold	90	3	0	0	4.0	19.0	0	140	0	1	0.67	74.47
Shredded Wheat spoon size	N	cold	90	3	0	0	3.0	20.0	0	120	0	1	0.67	72.80
Smacks	K	cold	110	2	1	70	1.0	9.0	15	40	25	2	0.75	31.23
Special K	K	cold	110	6	0	230	1.0	16.0	3	55	25	1	1.00	53.13
Strawberry Fruit Wheats	N	cold	90	2	0	15	3.0	15.0	5	90	25	2	1.00	59.36
Total Corn Flakes	G	cold	110	2	1	200	0.0	21.0	3	35	100	3	1.00	38.84

Brand	Manuf.	Type	Calories	Protein (g)	Fat (g)	Sodium (mg)	Fiber (g)	Carbo. (g)	Added Sugars (g)	Potassium (g)	Vitamins	Shelf Number	Cups Per Serv.	Customer Rating
Total Raisin Bran	G	cold	140	3	1	190	4.0	15.0	14	230	100	3	1.00	28.59
Total Whole Grain	G	cold	100	3	1	200	3.0	16.0	3	110	100	3	1.00	46.66
Triples	G	cold	110	2	1	250	0.0	21.0	3	60	25	3	0.75	39.11
Trix	G	cold	110	1	1	140	0.0	13.0	12	25	25	2	1.00	27.75
Wheat Chex	R	cold	100	3	1	230	3.0	17.0	3	115	25	1	0.67	49.79
Wheaties	G	cold	100	3	1	200	3.0	17.0	3	110	25	1	1.00	51.59
Wheaties Honey Gold	G	cold	110	2	1	200	1.0	16.0	8	60	25	1	0.75	36.19

The final column, *Customer Rating*, represents the percentage of customers who gave a favorable rating to each cereal in consumer reports. Now consider the question, *"Do sugary cereals tend to receive higher ratings?"*. To answer this question, we will fit a linear model using the variable *Added Sugar* as a predictor of *Customer Rating*. The fitted linear model is displayed in the plot below.



In panel $\mathbf{A}$ above, the red line indicates the least-squares regression line. The red dots give the mean of the distribution of customer ratings for each value of added sugars. As is illustrated in plot, the least squares model relates the mean of each conditional distribution $y|x$ across all values of the predictor. Panel $\mathbf{B}$ gives the boxplots for each conditional distribution.

- Do you believe there's a significant linear relationship between customer rating and the amount of added sugars?
- What does the fitted linear model indicate about the relationship between customer rating and added sugar?

Some summary statistics regarding each variable is given in the table below

Statistic	Avg. Customer Rating	Avg. Added Sugars (g)	StDev Customer Rating	StDev. Added Sugars (g)	$r_{x,y}$	Total SS	Regression SS	Error SS	$SE(\hat{\beta})$	$SE(\hat{\alpha})$
Value	42.56	7.03	14.11	4.38	-0.76	14929.29	8711.93	6217.36	0.24	2.00

- Using the table above, compute the *intercept*, *slope*, and r^2 for the regression model treating customer rating as the response and interpret them

$$\hat{\beta} = -0.76\left(\frac{14.11}{4.38}\right) \approx -2.46$$

$$\hat{\alpha} = 42.56 - (-2.46) \cdot 7.03 = 59.85$$

$$r^2 = \frac{8711.93}{14929.29} = (-0.76)^2 = 0.58 \text{ or } 58\%$$

- Test the hypothesis that the slope is significantly different than zero and interpret your results

The null and alternative hypotheses are

$$H_0 : \beta = 0, \quad H_A : \beta \neq 0$$

The test statistic is

$$t_0 = \frac{-2.46}{0.24} = -10.25 \sim t(76 - 2)$$

The p-value is

$$P(|t| \geq |t_0||H_0 \text{ True}) = 2P(t \geq 10.25|H_0 \text{ True}) \approx 0$$

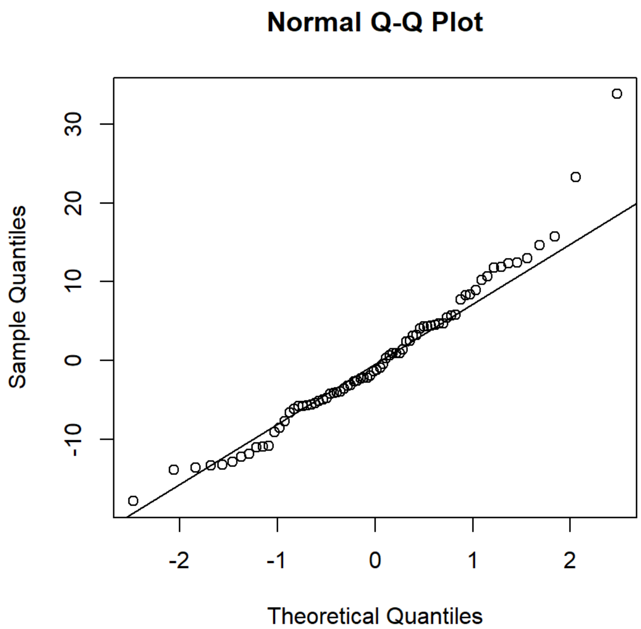
- Compute the F-statistic for the ANOVA test using the table above:

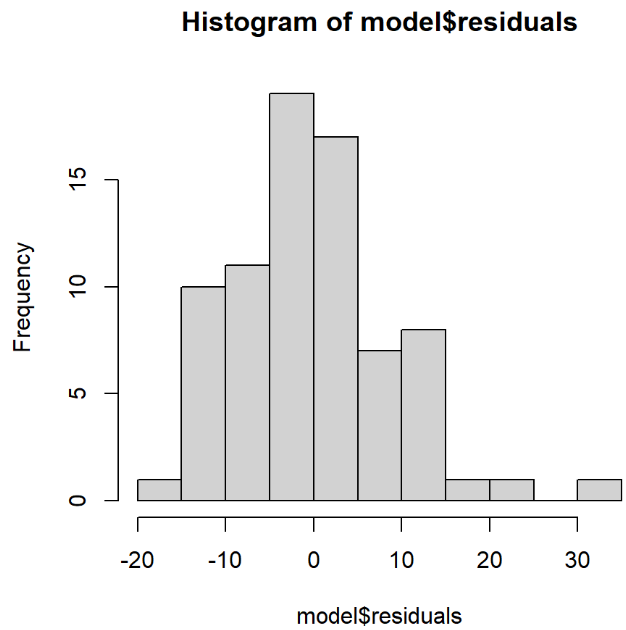
$$F_0 = \frac{8711.93/1}{6217.36/(76 - 2)} = 103.69 \sim F(1, 74)$$

Checking Assumptions

Checking assumptions in simple linear regression is crucial because it ensures the validity and reliability of your model's predictions and conclusions. Recall that the least-squares estimator makes several strong assumptions about the structure of the data (*linearity*, *independence*, *constant variance*, and *normality* of residuals). If these assumptions are violated, it can lead to biased estimates of the slope and intercept, incorrect inferences, or unreliable predictions. Our estimates for the slope and intercept can also be biased by influential points such as outliers in the data. Thus, when fitting the least-squares model, it is best to pair results with a thorough investigation of the residuals and any potential outliers.

A common plot used to check the assumptions of residuals in linear regression is the quantile-quantile, or "Q-Q," plot. This plot compares the distribution of the computed residuals to the theoretical quantiles of a standard normal distribution. Below is the Q-Q plot and histogram for the residuals of the linear regression model fitting customer ratings as a response of added sugar content:





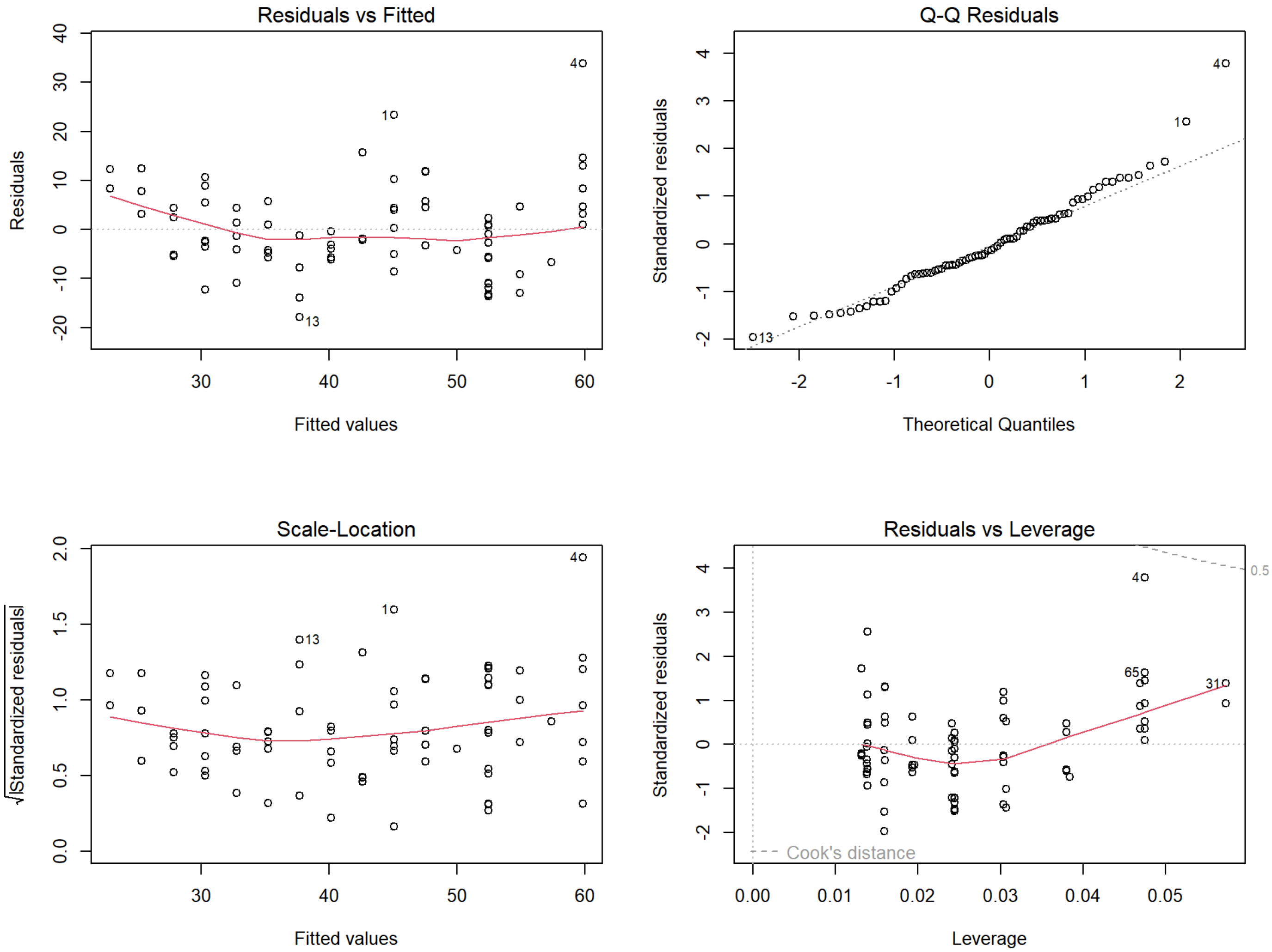
If the residuals are approximately normally distributed, the points on the Q-Q plot will align along a diagonal line, indicating a one-to-one correspondence with the standard normal quantiles. However, if the residuals deviate from this diagonal, especially at the ends, it may indicate skewness or other departures from normality. Outliers and extreme values in the residuals will appear as points that are significantly distant from the diagonal line.

- If residuals are non-normal, often rescaling the response y with a *non-linear* transformation such as taking the $\log$ or square-root can help

Another common plot is called a "residual-leverage" plot. *Leverage* is a measure that can help assess the influence a data point has on the regression line. A high leverage value indicates that the data point has a greater potential to affect the fit of the regression model. Often the residuals are given as the *standardized residuals* (denoted ϵ_i^*) - the raw residuals divided by the estimated residual standard deviation $\hat{\sigma}_\epsilon$.

$$\hat{\epsilon_i}^* = \frac{\hat{\epsilon}_i}{\hat{\sigma_\epsilon}}$$

Residuals are typically standardized to help improve the ability to identify outliers - especially if the raw residuals have non-constant variance. Software like R makes it easy to quickly produce several diagnostic plots to assess assumptions and outliers. The residual-leverage plot for the cereal example is shown below:



In the four plots above, the residuals vs leverage plot is given in the bottom-right panel. The x-axis is *leverage* or influence that each data point has on the regression line. This plot also features something called **Cook's Distance** given by the dashed lines on the plot. Cook's distance is another metric to assess the influence of individual points. Generally speaking, points that fall above or below the dashed lines (i.e the Cook's distance) are potential outliers. As we can see from the residual vs leverage plot above, the data point labeled "4" is close but not above the dashed line indicating there are no outliers. Another example of a residual vs leverage plot and how to address potential outliers can be found here https://www.statology.org/residuals-vs-leverage-plot/ (https://www.statology.org/residuals-vs-leverage-plot/)

The last two plots we will discuss are the *Scale-Location plot* (bottom left panel in the figure above) and the *Residuals vs Fitted plot* (top left panel in the figure above). These two plots are crucial for assessing the constant variance assumption of the residuals in a linear model.

In both plots, the x-axis represents the predicted values $\hat{y}_i$, while the y-axis represents either the residuals (for the Residuals vs Fitted plot) or the scaled residuals (for the Scale-Location plot). The objective of these plots is to check whether the residuals maintain a constant variance as the predicted values change.

In the Residuals vs Fitted plot, the residuals should form a cloud-like shape around the horizontal line representing zero. If the red line, which represents a smooth trend through the residuals, is roughly horizontal, this suggests that the variance is constant across the predicted values. However, if the red line exhibits a discernible pattern, such as a curve or slope, this could indicate non-constant variance.

Similarly, in the Scale-Location plot, the y-axis represents the square root of the absolute standardized residuals, and the red line shows the trend of these scaled residuals. The interpretation is the same as it is for the Residuals vs Fitted plot, however scaling the residuals in this way can make it easier to visually identify patterns that might suggest non-constant variance. You can see another example of these two plot types here https://www.statology.org/scale-location-plot/ (https://www.statology.org/scale-location-plot/)

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